1. Prerequisites

A knowledge of the specification for P1 and P2, their prerequisites and associated formulae, is assumed and may be tested.

It is also necessary for students:

- to have a knowledge of location of roots of $f(x) = 0$ by considering changes of sign of $f(x)$ in an interval in which
- $f(x)$ is continuous
- to have a knowledge of rotating shapes through any angle about (0, 0)
- to be able to divide a cubic polynomial by a quadratic polynomial
- to be able to divide a quartic polynomial by a quadratic polynomial.

2. Examination

- First assessment: June 2019.
- The assessment is 1 hour and 30 minutes.
- The assessment is out of 75 marks.
- Students must answer all questions.
- Calculators may be used in the examination. Please see *Appendix 6: Use of calculators*.
- The booklet *Mathematical Formulae and Statistical Tables* will be provided for use in the assessments.

3. Notation and formulae

Students will be expected to understand the symbols outlined in *Appendix 7: Notation*.

Formulae that students are expected to know are given below and will **not** appear in the booklet *Mathematical Formulae and Statistical Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

This is a list of formulae that students are expected to remember and which will not be included in formulae booklets.

Roots of quadratic equations

For $ax^2 + bx + c = 0$: $\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$

Series

$$\sum_{r=1}^{n} r = \tfrac{1}{2}n(n+1)$$

FP1.3 Unit content

What students need to learn:		Guidance
1. Complex numbers		
1.1	Definition of complex numbers in the form $a + \mathrm{i}b$ and $r\cos\theta + \mathrm{i}\,r\sin\theta$.	The meaning of conjugate, modulus, argument, real part, imaginary part and equality of complex numbers should be known.
1.2	Sum, product and quotient of complex numbers.	$\lvert z_1 z_2\rvert = \lvert z_1\rvert\,\lvert z_2\rvert$ Knowledge of the result $\arg(z_1 z_2) = \arg z_1 + \arg z_2$ is not required.
1.3	Geometrical representation of complex numbers in the Argand diagram. Geometrical representation of sums, products and quotients of complex numbers.	
1.4	Complex solutions of quadratic equations with real coefficients.	
1.5	Finding conjugate complex roots and a real root of a cubic equation with integer coefficients.	Knowledge that if z_1 is a root of $f(z) = 0$ then z_1^* is also a root.
1.6	Finding conjugate complex roots and/or real roots of a quartic equation with real coefficients.	For example, (i) $f(x) = x^4 - x^3 - 5x^2 + 7x + 10$ Given that $x = 2 + \mathrm{i}$ is a root of $f(x) = 0$, use algebra to find the three other roots of $f(x) = 0$ (ii) $g(x) = x^4 - x^3 + 6x^2 + 14x - 20$ Given $g(1) = 0$ and $g(-2) = 0$, use algebra to solve $g(x) = 0$ completely.